



Game Production – Overview & Intro

MTD.ba/DA.ba SS 2026, Hagenberg

John Kulha

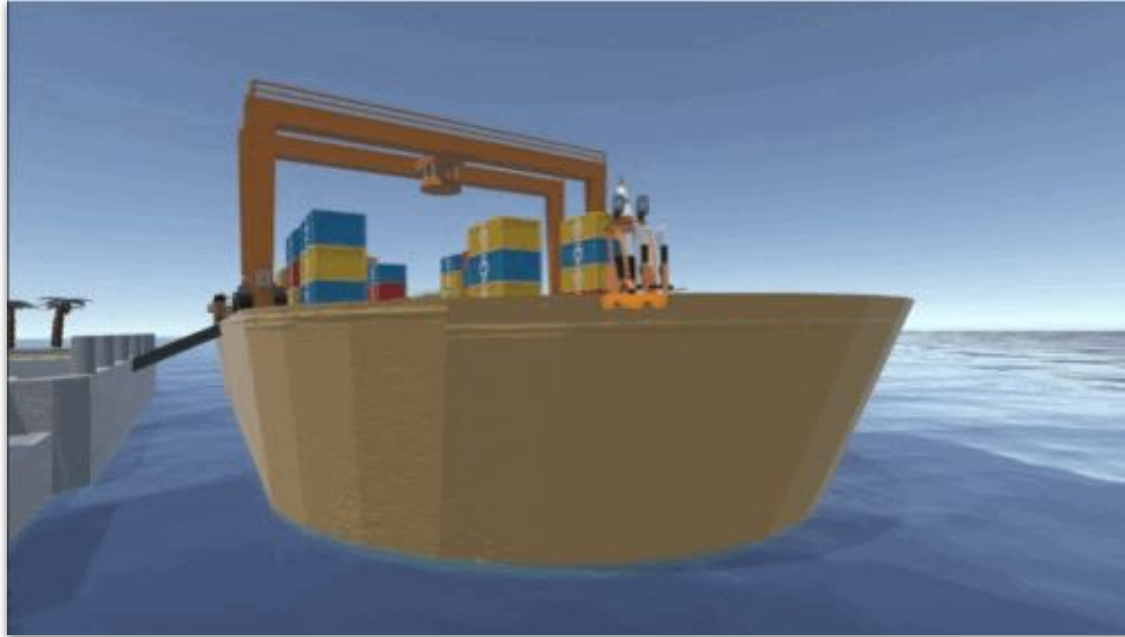
Unity Developer, Programmer

- > Informationstechnologie
- > Game Engineering @ FH Technikum
- > Indie Dev @ twonodes
- > Programmer, Game Designer, ...
- > Teaching @FH-Hagenberg,
Technikum & HTL Spengergasse

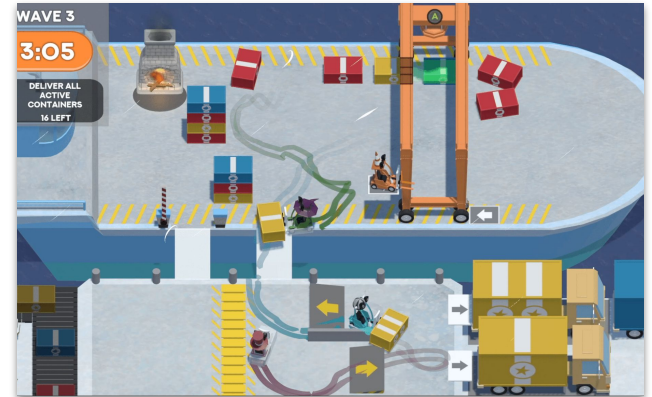
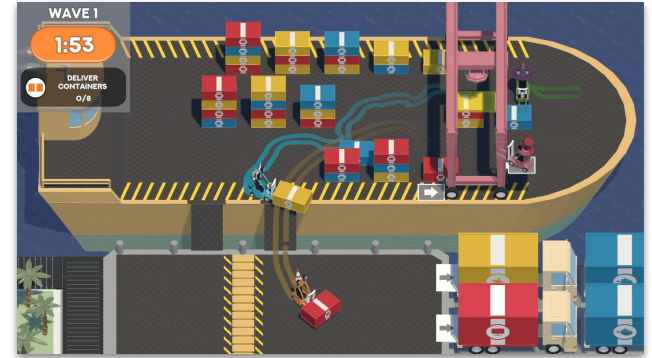
john.kulha@fh-hagenberg.at

Discord: antistreber





Container Masters





Super Mall Tycoon

Mit 14 verliebt – 12 Jahre später steht HTL-Lehrerpaar vor Marktreife mit Gaming-Startup

Wie man ein Spiel zur Marktreife führt, das erklären John und Beate Kulha. Die beiden sind seit ihrer Kindheit ein Paar und entwickeln Computerspiele abseits des Mainstreams - alle mit einem Touch von Management-Skills.

Artikel Link kopieren



(c) zVg - John und Beate Kulha von Twonodes.

Mit jungen 14 Jahren haben sie einander gefunden. Es dauerte einen Monat in der neuen Klasse einer Wiener HTL, bis sie ein Paar wurden und ihre Reise begann. Sie führte sie zum Abschluss der technischen Lehranstalt, durchs Studium, zur gemeinsamen (Teilzeit-)Lehrerschaft an einer anderen Wiener HTL und im Hintergrund stets parallel mitlaufend zur Computer-Spieleentwicklung. Beate und John Kulha haben stetig Spiel-Ideen im Kopf, suchen dabei gezielt Nischen aus und sind dabei, ihr Startup Twonodes zu gründen.

Twonodes und Super Mall Tycoon

„Uns war immer klar, dass wir uns selbstständig machen wollen“, sagt John. „Und wir haben bereits zu Schulzeiten gemerkt, dass man mit Programmieren coole Sachen machen kann.“

Zwölf Jahre später ist einiges geschehen und noch mehr scheint bevorzustehen. Das Gründerpaar hat im Vorjahr für ihr Spiel „Container Masters“ (gemeinsam mit Sebastian Hofer und Lukas Weiss als Masterarbeit) den „Best of Class“-Award an der FH-Technikum Wien gewonnen und durfte als Preis das Spiel auf der Spielemesse Gamescom in Köln vorstellen.

Bei diesem Logistik-Koop-Game geht es darum, Schiffe zu entleeren und die Container zu LKWs zu liefern. Mitspieler:innen können dabei Chaos stiften und blockieren, erklärt Beate.

Auf der **Aninite** des heurigen Sommers präsentierten die Kulhas indes ihr neuestes Werk „Super Mall Tycoon“, wo Spieler:innen ein Einkaufszentrum managen, inklusive Vermietung von Räumen, Ausbau, Überblick über Rentabilität sowie ROI vonseiten der Mieter, Parkplatzmöglichkeiten und Kund:innenwünsche.

„Wir haben uns diese spezifischen Bereiche ausgesucht, weil es 'Nischen' sind“, erklärt John Kulha. „Spiele bei denen man nachdenken muss. Das wird immer seltener, wir glauben aber, dass



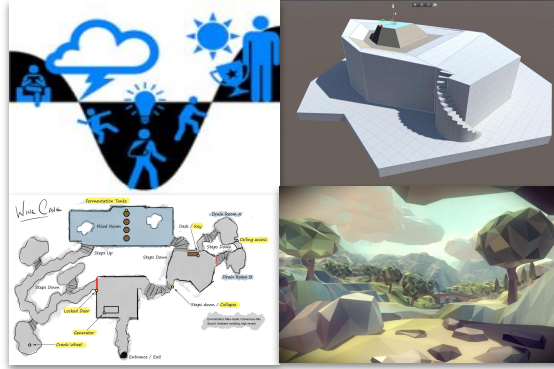
Game Production

MTD + DA

Goals of this Course



Working with
Unity 3D



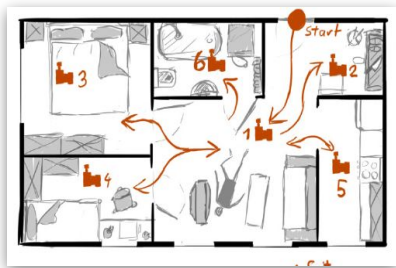
Introduction to
Game Production



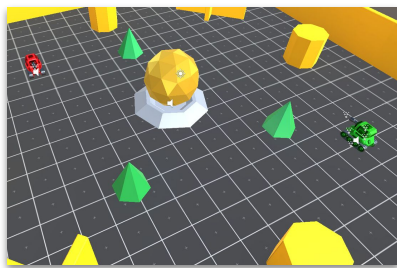
Introduction to
Working in
Teams

Goals of this Course

Game Prototype



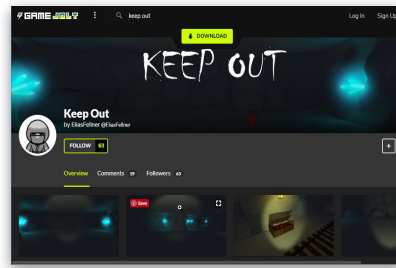
Concept



Gameplay
Prototype



Level
Prototype



Publishing

Game Constraints

Minimal constraints, but scalable!

- 5-15 min, single player, standalone, PC & Mac
- 3D game with 1st or 3rd person controller
- *Simple gameplay...*
- Only prototyping or low-poly assets
- Simple or no UI
- **Teamwork** (4-6 ppl)



“Simple Gameplay”

We recommend (& will provide basics) for **story based adventure gameplay**:

Simple 1st or 3rd person interaction
(*walk, look, pick up, listen to, read*), focus on
storytelling and level design.

*Think: Gone Home, Firewatch, Edith Finch,
Amnesia, Dear Esther, The Stanley Parable,...*



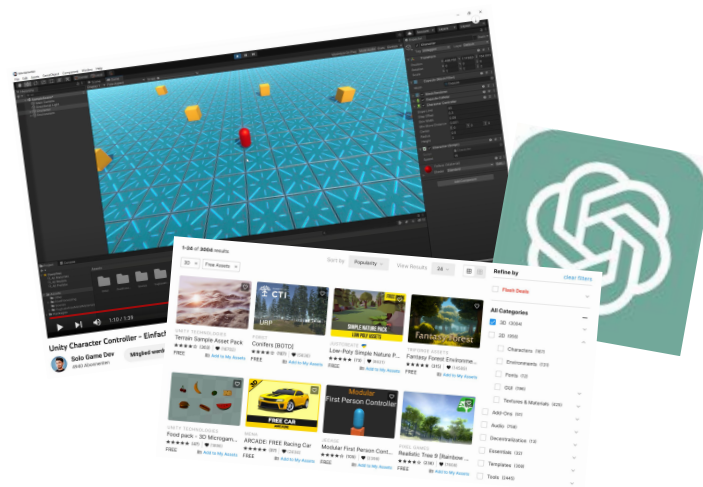


Custom Gameplay

Custom gameplay is a programming challenge & we can only cover the basics this semester 😞

You may create custom mechanics, but **expect more work!**

If you do, we recommend using **tutorials, 3rd party assets, Copilot, etc** and will help you choose the right ones!



GitHub Student Developer Pack

Learn to ship software like a pro. There's no substitute for hands-on experience. But for most students, real world tools can be cost-prohibitive. That's why we created the GitHub Student Developer Pack with some of our partners and friends.

[Sign up for Student Developer Pack](#)

Love the pack? Spread the word



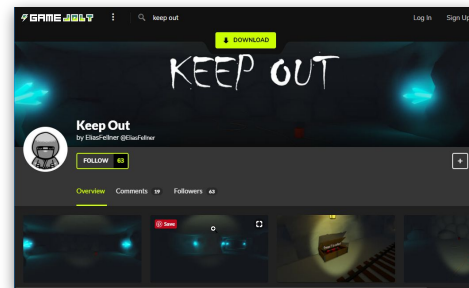
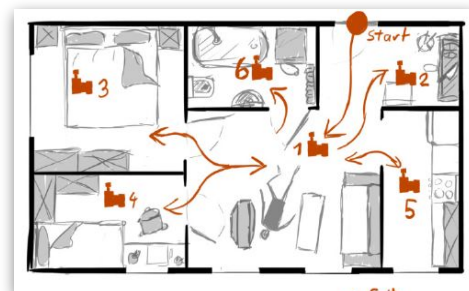
Experiences

Discover the best ways to use pack offers with Experiences. Experiences are curated bundles of pack partner products, GitHub tools, and other resources that are designed for you learn new skills and make the most out of the Student Developer Pack and your journey in GitHub Education.



Steps

- > **Game Concept**
Team & roles, gameplay, story, progression, project planning, documentation
- > **Prototype**
Basic gameplay, controllers, UI, camera, iterating
- > **Level Greyboxing**
Layout, progression, leading players, teaching gameplay through levels, iterating
- > **Content Creation**
Custom assets, audio, lighting
- > **Playtesting**
- > **Publishing**
MTD/DA Games platforms, build, presentation page itch.io
- > **Development Journal**
Regular updates during the production



Grading

Milestones

Moodle Submissions, Presentations, Builds, etc.

1. Pitch Video 6.25%
- 2. Concept 18.75%**
3. Project Plan 12.5%
4. Work in Progress 12.5%
5. Playtest 12.5%
6. Devlogs & PR Page 12.5%
- 7. Final Game 25%**

→ Requirements met, progress visible, presentation held,...

Max 9 Points

0-1,99 Points: Nicht Genügend
2-3,99 Points: Genügend
4-5,99 Points: Befriedigend
6-7,99 Points: Gut
8-9 Points: Sehr Gut

Groups!

Each person in a group gets the same points, unless requested otherwise!

Livestream



Topics

- > Game production
Conception basics, documentation, phases, roles, level design
- > Project Management
Agile PM with SCRUM and Trello, estimating
- > Working in teams
Communication, version control, presenting
- > Unity
Vibe coding, simple asset workflows, building

Not covered:

Game design, game art, asset creation (modelling, audio production,...), in depth programming

Tools

Teams (Calls, Questions,...)

Moodle (Slides, Files, Assignments)

<https://elearning.fh-ooe.at/course/view.php?id=28087>

GitHub (Version Control, t.b.a.)

Hagenberg Game Wiki (Documentation)

<https://hagenberg.games/wiki/Hauptseite>

Hagenberg Hub (Dev-Logs & Presskits)

<https://hub.hagenberg.games/user/auth/login>

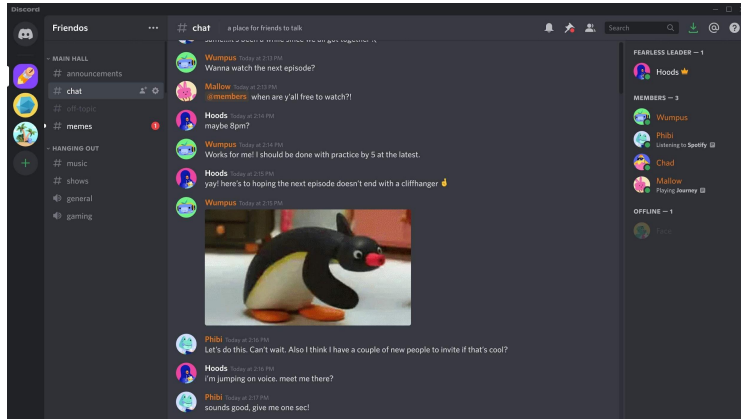
The screenshot shows the Moodle course page for 'GAP2IL: Game Production (DA.ba VZ SS23)'. The page includes a navigation bar with links to 'KURSE', 'SERVICES FH OÖ', 'SUPPORT', 'TOP Lehre', and 'Deutsch (de)'. The course title is 'GAP2IL: Game Production (DA.ba VZ SS23)' with a 'Bearbeiten ausschalten' button. The 'Overview' section describes the course as a small video game prototype with a 10-15 min playtime, covering development phases from concept to production. The 'Topics' section lists: Game Concepts, Game Design Documents, Production, Play Testing, Version Control, Game Dev Roles, Project Management, Research, Presenting, Unity Development, C# Scripting, Game PR. The 'Grading' section shows a scale from 0-8 (Not Enough) to 15-16 (Out). The 'Kalender' (Calendar) shows dates for March 2023. The 'Aktuelle Termine' (Current Dates) section lists dates for various events. The 'Tools & Links' section is at the bottom.

The screenshot shows the 'Game Production Ressourcen' page on the Hagenberg Game Wiki. The page includes a navigation bar with links to 'Seite', 'Diskussion', 'Lesen', 'Bearbeiten', 'Qualität bearbeiten', 'Versionsgeschichte', 'Wählen', and 'hagenberg games Wiki durchsuchen'. The 'Hauptseite' (Main Page) section lists: Letzte Änderungen, Zählige Seite, Hilfe zu Modulen, Werkzeuge, Links auf diese Seite, Änderungen an verlinkten Seiten, Diese Vorhaben, Sprechblumen, Druckversion, Permanenter Link, Seiteninformationen, Seite zitieren. The 'Inhaltsverzeichnis' (Table of Contents) lists: 1. Game Vision, 2. Game Design, 3. Storytelling, 4. Level Design, 5. Project Management, 6. Free Low Poly Models, 7. PR & Communication. The 'Game Vision' section includes a table with columns: Titel, Beschreibung, Autor*in, and Link. The 'Game Design' section includes a table with columns: Titel, Beschreibung, Autor*in, and Link.

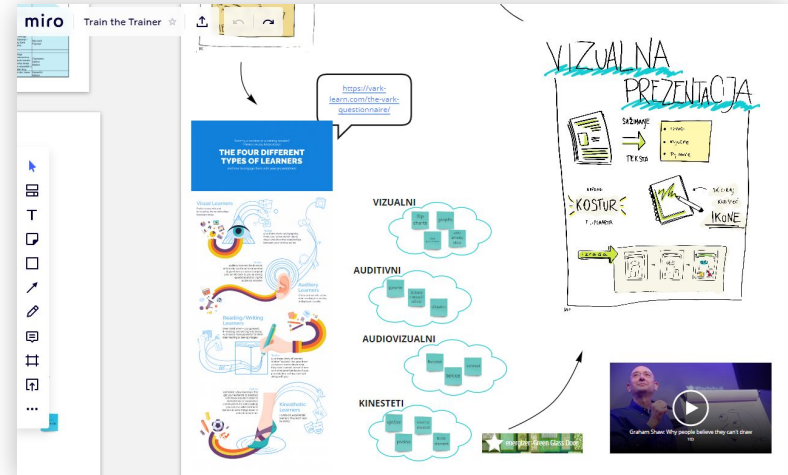
Titel	Beschreibung	Autor*in	Link
Game Design Vision Statements	Kurze Beschreibung zum Thema "Game Vision"	Gregory Avery/Weir (Future Proof Games)	https://futureproofgames.com/blog/2020/02/11/game-design-vision-statements/#

Titel	Beschreibung	Autor*in	Link
See How Others Built Their Games!	Game Design Dokumente verschiedener Spiele (Diablo, DOOM (Böbe, Silent Hill 2 usw.)	Gamescroye	https://gamescroye.com/resources/game-design-documents/#
Next-Class Game Design	Game Design Dokument für Grand Theft Auto aus dem Jahr 1995.	DMA Design Ltd	http://classes.dma.ucla.edu/Winter12/1574/doc2/gtda_gdd.pdf
From the Vault - The BioShock Pitch	Teil 1 des originalen BioShock Pitch Dokumente	Irrational	https://irrationalgames.ghoststorygames.com/insider/from-the-vault-may/#
From the Vault - Part Two of the BioShock Pitch	Teil 2 des originalen BioShock Pitch Dokumente	Irrational	https://irrationalgames.ghoststorygames.com/insider/from-the-vault-june/#

Create your own infrastructure!



Text, voice & video calls



Collaborative tools

etc.

Important Dates

- > 22.5. - WIP Session
- > 12.6. - Playtest
- > 1.7. - Playtest x Spengergasse
- > 9.7. - Final Deadline!
- > **24.7. - Twitch Stream**



Tutors

Edina Abazovic, MTD (s2410238002@students.fh-hagenberg.at)

Amalia Pintilie, DA (s2410872023@students.fh-hagenberg.at)

Producing Games

BUILD A VIDEO GAME



Where to start?



Design

Mechanics & Dynamics
Level Design
Story & World Building
Character Design

...

Game Design Document
Concept Art, Mood Boards
Level, Progress, Pacing
Diagrams

...

Produktion

Project Phases & Workflows
Teamwork & Organisation
Testing
PR & Communication

...

Project Plans, Tasks
Standups, Presentations
Repositories
PR Pages

...

Tooling

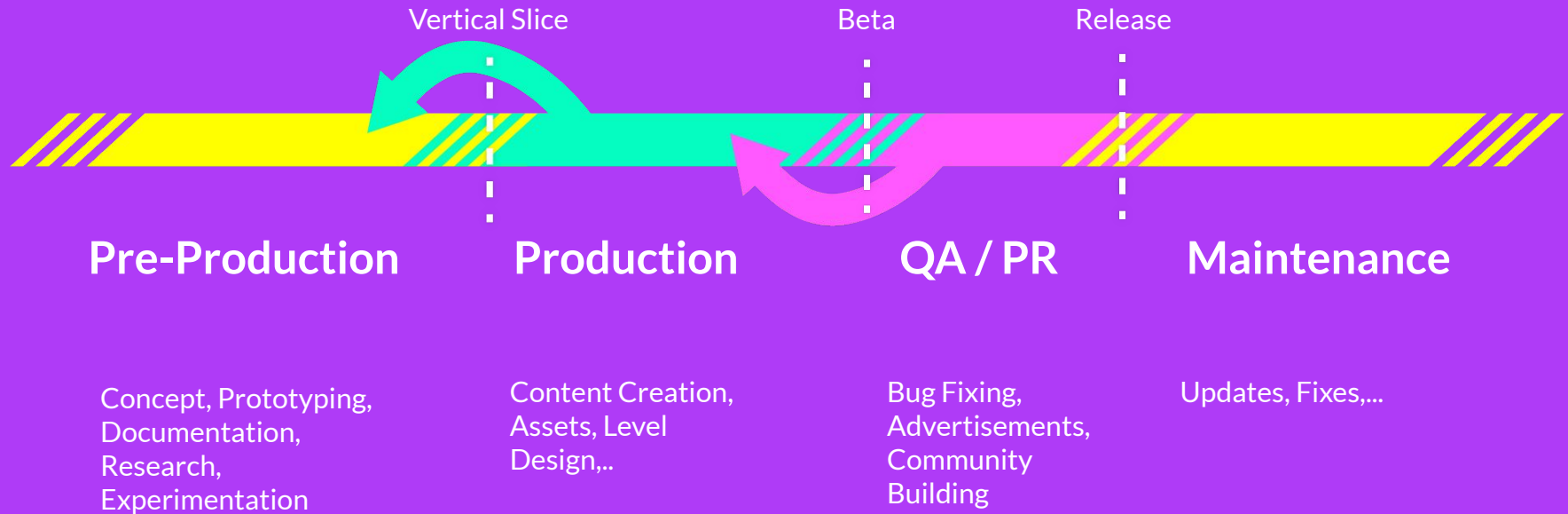
Modelling
Texturing
Programming
Graphics

...

Maya/Blender,
Unity/Unreal,
Game Programming,
Computer Graphics,

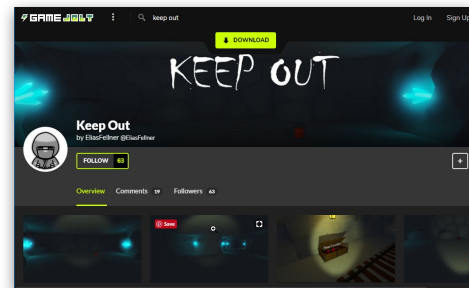
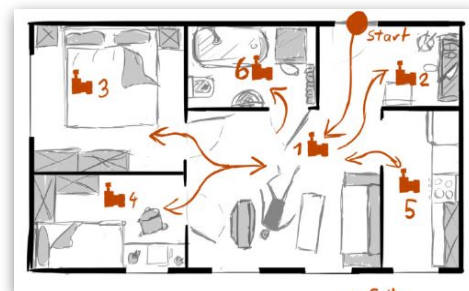
...

Example Life-Cycle of a Small to Mid Game...



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Topic:
**Project Management &
Communication**

Programming

Modelling

VFX

Sounds

Music

Environment Art

Character Animation

Publishing

Texturing

Level Design

Story

Testing

Rigging

Advertisement

Localisation

System Programming

Engine Programming

GUI Design

Platform

Game Design

Character Art

Gameplay
Programming

Programming



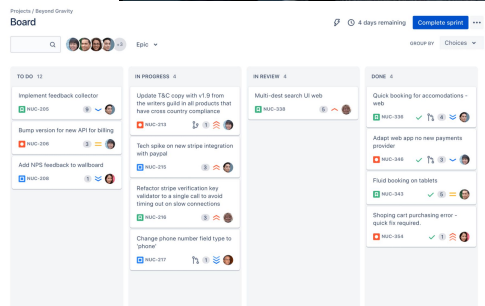
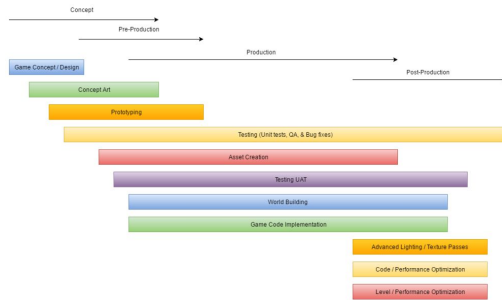


Spelunky, Derek Yu



Assassin's Creed: Revelation, Ubisoft Montreal
(4000+ employees)

How to coordinate all people in a team?



SCRUM as one framework for Game Production

- Tasks & Packages
 - Tasks, Stories & Epics
 - Task-Status (Backlog, Todo, In Progress,...)
- Time Estimations
 - Story Points
- Phases
 - Sprints
- Communication
 - Daily Scrum, Sprint Review,...
- Roles
 - Project Owner, Scrum Master,...
- Analysis & Overview
 - Burndown Charts,...

SCRUM

“Agile” project management method

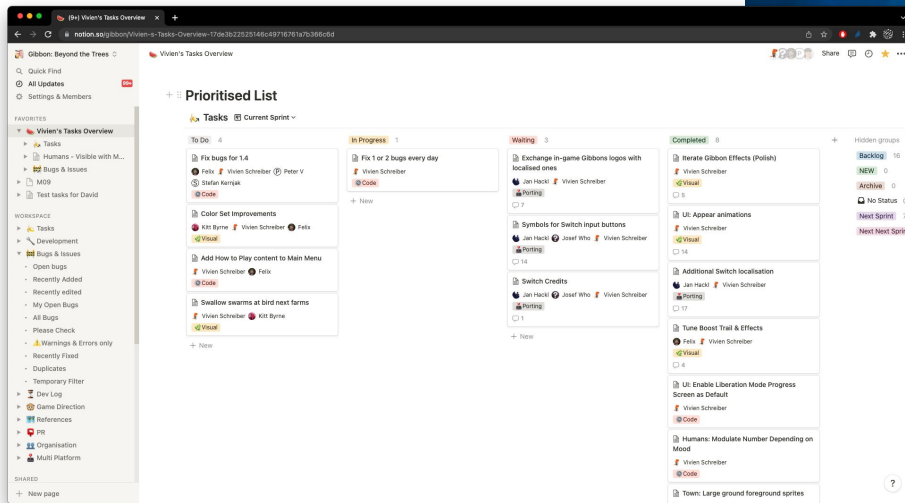
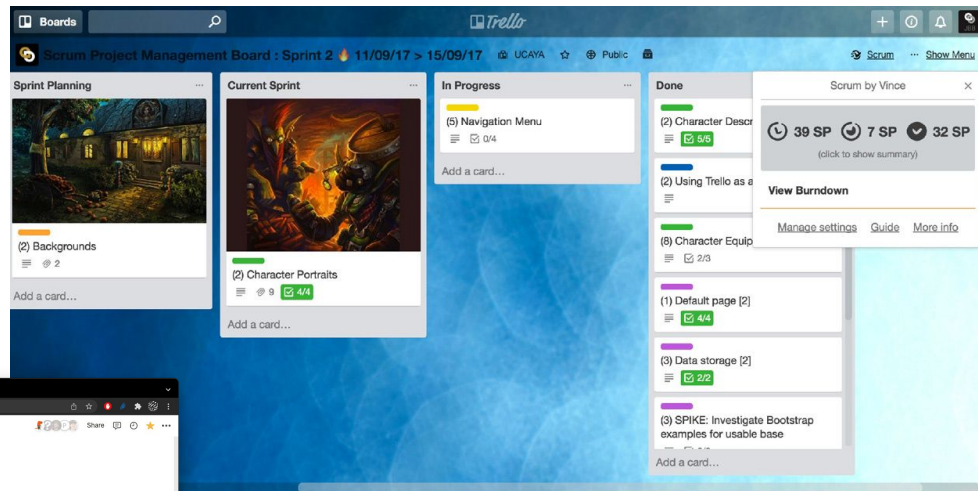
- > Teams of 3-9 members
- > Project is broken down into smaller tasks (“**User Stories**”)
- > Tasks are placed on a **board**
- > Tasks progress through states until finished
- > Work is structured in iterations of 1-3 weeks (“**Sprints**”)

Continuous developing, testing & releasing of playable versions



Stories	To Do		In Progress	Testing	Done
This is a sample text. Replace it with your own text.	This is a sample text. Replace it with your own text.	This is a sample text. Replace it with your own text.	This is a sample text. This is a sample text. This is a sample text.	This is a sample text. This is a sample text. This is a sample text.	This is a sample text. Replace it with your own text. This is a sample text. Replace it with your own text.
This is a sample text. Replace it with your own text.	This is a sample text. This is a sample text.	This is a sample text. This is a sample text.	This is a sample text. This is a sample text.	This is a sample text.	This is a sample text. Replace it with your own text.

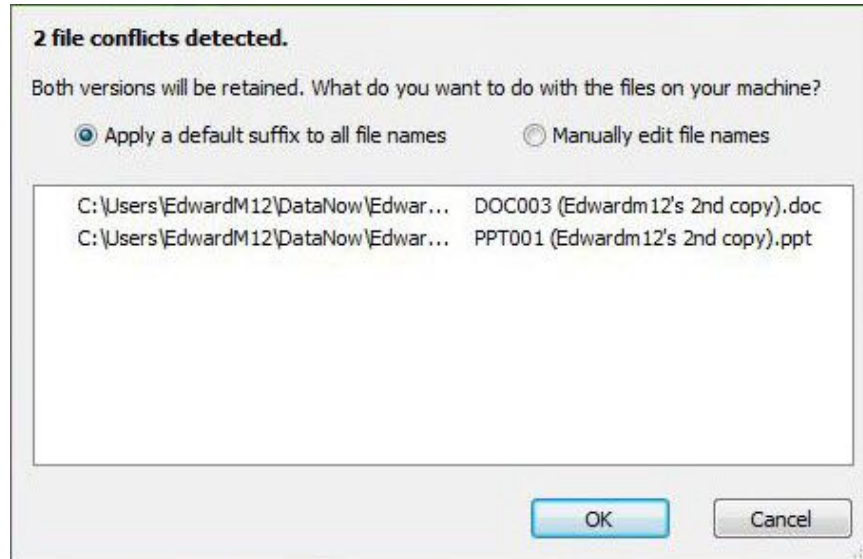
Scrum-ish



Source Control



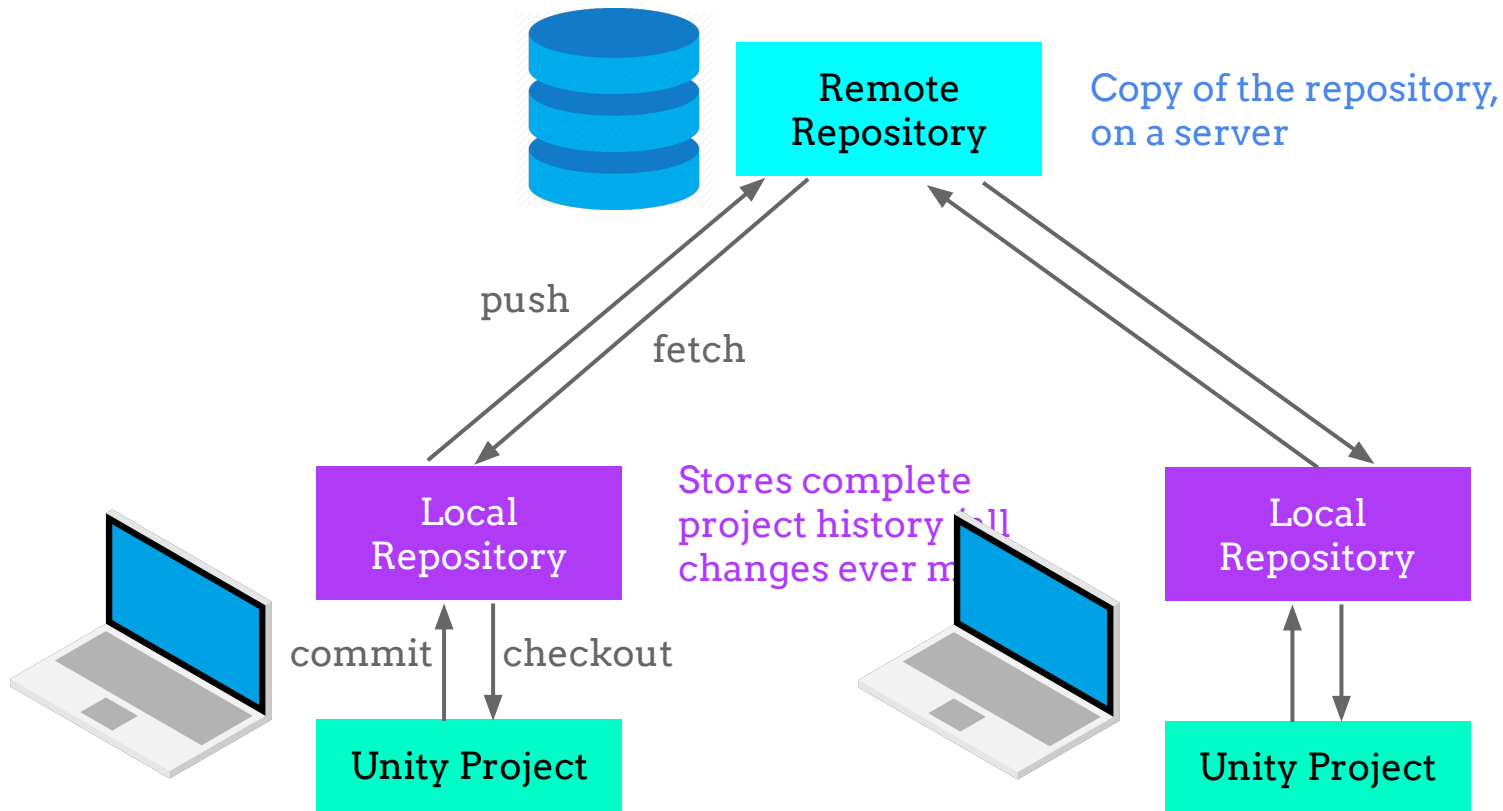
Problems when working on digital projects with teams...



Name
Press release for approval.doc
Press release final.doc
Press release FINAL VERSION.doc
Press release FINAL FINAL VERSION.doc
IMPROVED FINAL PRESS RELEASE.doc
REVISED APPROVED FINAL PRESS RELEASE.doc
REVISED APPROVED FINAL PRESS RELEASE v. 2.doc
!! NEW REVISED APPROVED FINAL PRESS RELEASE v. 2.doc
!!! REVISED NEW REVISED APPROVED FINAL PRESS RELEASE v. 2.doc
!!!! Press release as sent.doc

Version Control & GIT





Not 100% accurate,
but good enough for now!

Git

One of several professional version control systems

- > Distributed = whole project history lies on every computer
- > Open source & free
- > Widely used in the games industry
- > Different servers available (GitHub, GitLab, BitBucket,...)]

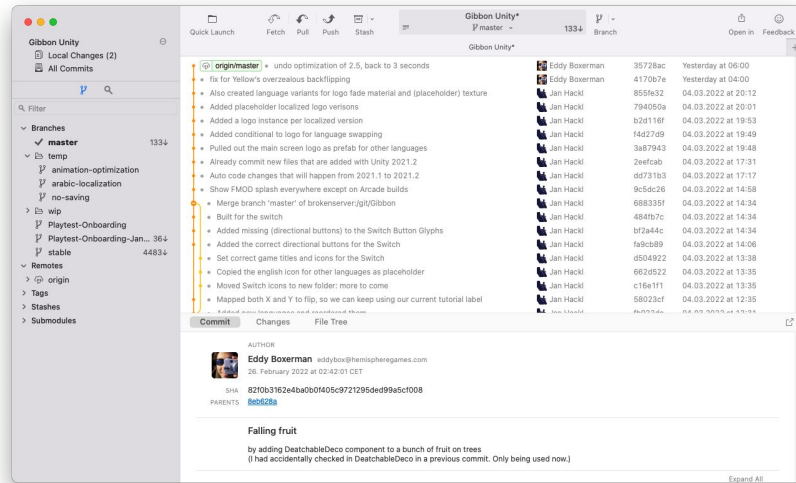
What you get:

- > Version history
- > Conflict management
- > Backups

Other version control systems: SVN, Mercurial, Perforce,...

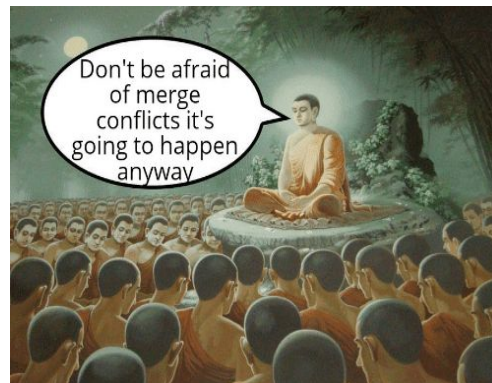
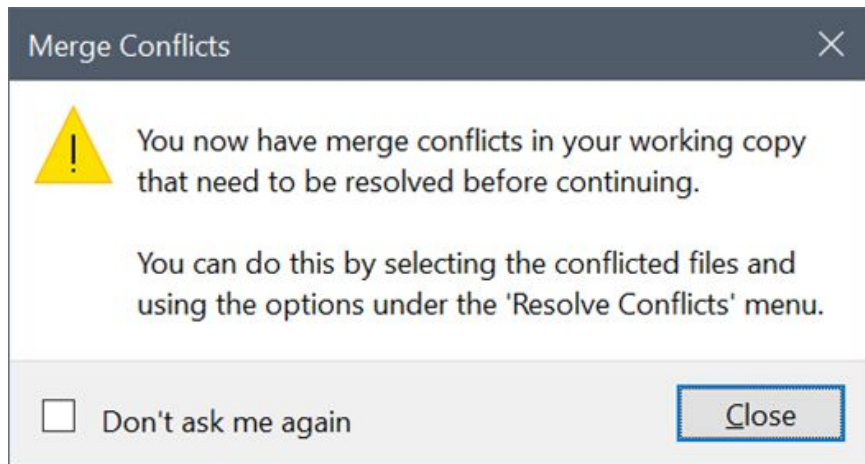
```
caelj@mac.local ~ $ mkdir repo
caelj@mac.local ~ $ cd repo
caelj@mac.local repo $ git init .
Initialized empty Git repository in /Users/caelj/repo/.git/
caelj@mac.local repo $ touch foo
caelj@mac.local repo $ git stage .
caelj@mac.local repo $ git commit -m "Epoch."
[master (root-commit) 9ed3c75] Epoch.
 0 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 foo
caelj@mac.local repo (master) $ git checkout -b some_feature_branch
Switched to a new branch 'some_feature_branch'
caelj@mac.local repo (some_feature_branch) $ git branch
  master
* some_feature_branch
caelj@mac.local repo (some_feature_branch) $ git checkout master
Switched to branch 'master'
caelj@mac.local repo (master) $
```

Git is a **command-line tool**

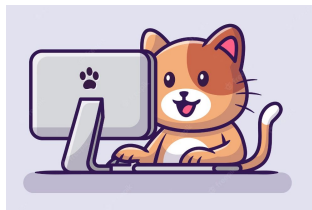


We use a **GUI client** for comfort (eg Source Tree, Fork, GitHub client,...)

Conflicts are not nice, but at least you notice them!



Being the git guy in a project...



Developer 1

Starts working on
level A backgrounds

Done with work,
Wants to push changes to
level A to repo

Conflict!

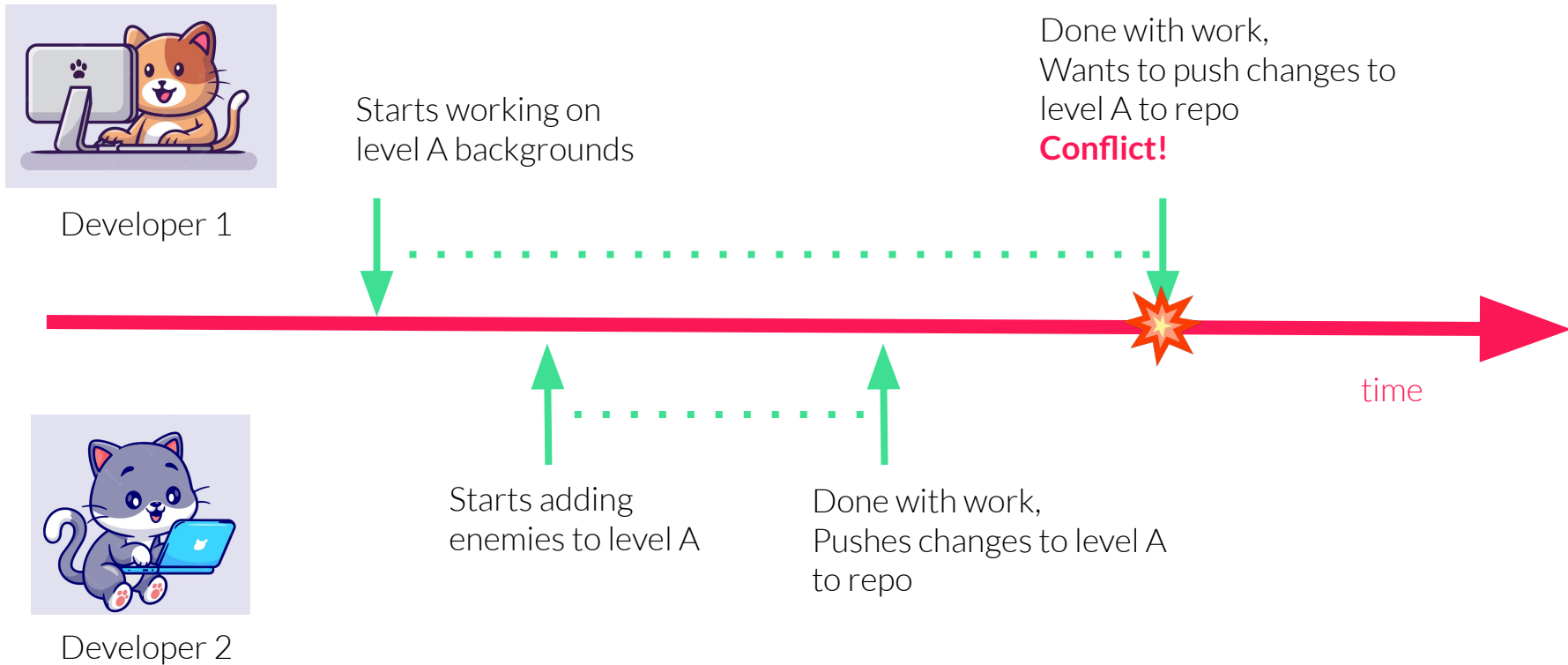


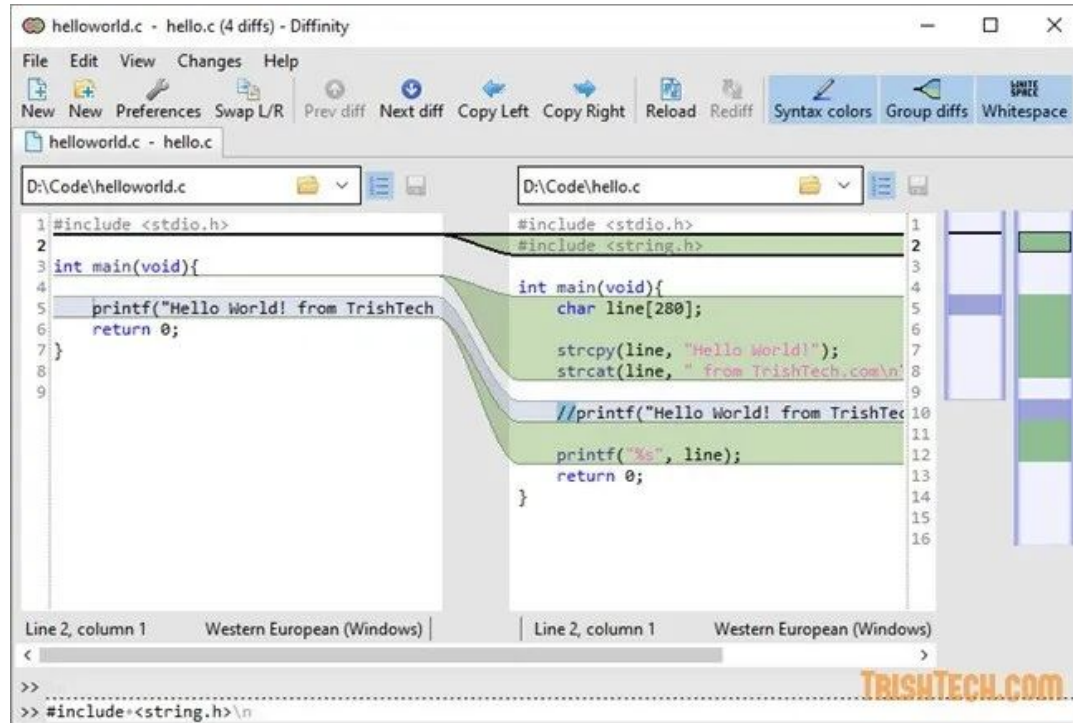
Developer 2

Starts adding
enemies to level A

Done with work,
Pushes changes to level A
to repo

time





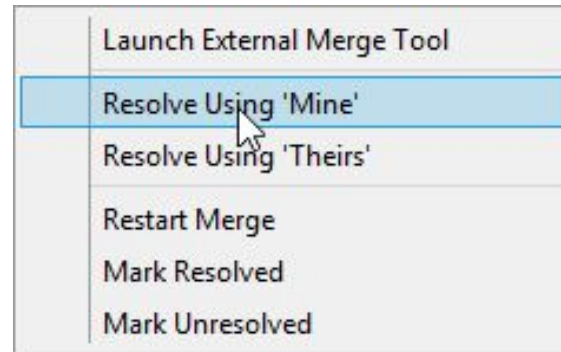
Conflicts in **code** can be resolved in detail with a **merge tool**.



W: 656px | H: 469px (284 KB)



W: 656px | H: 469px (295 KB)



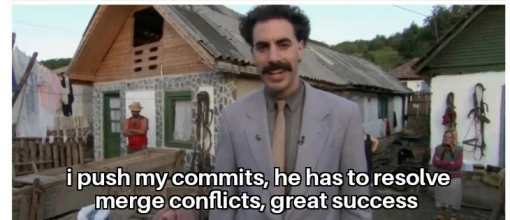
Conflicts in **binary files** (eg levels, images, audio,...) can only be resolved by selecting one or the other version...😞

“git”: British slang for a stupid or unpleasant person



I'm an egotistical bastard, and I name all my projects after myself. First Linux, now git.

(Linus Torvalds)





**Thank you &
See you today for first exercise!**



Title Graphic: Kenney Prototyping Textures

<https://github.com/Calinou/kenney-prototype-textures>



Play Sessions!